

Day 19 – Terraform & Infrastructure as Code (IaC)

Introduction

Infrastructure as Code automates cloud infrastructure using code.

Terraform

Terraform is an open-source IaC tool by HashiCorp.

Benefits

Automation, scalability, consistency, version control.

Components

Provider, Resource, Variables, State File, Modules.

Workflow

terraform init → terraform plan → terraform apply.

Real-World Project

Deploy VPC, EC2, and S3 automatically using Terraform.

Security

IAM permissions and secure Terraform state management.

Monitoring

CloudWatch tracks EC2 and infrastructure health.

Use Cases

DevOps automation, Kubernetes infrastructure, CI/CD.

Terraform Architecture Diagram

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